

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2026

Regulation: R20 | University: JNTUK

Design & Analysis of Algorithms (21CS51)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Asymptotic Analysis]

- a) Solve the given recurrence relation using the master theorem.
- b) Explain asymptotic notations with examples.

UNIT - 2 [Divide & Conquer and Greedy Methods]

- a) Solve the fractional knapsack problem using the greedy method.
- b) Solve the fractional knapsack problem using the greedy method.

UNIT - 3 [Dynamic Programming]

- a) Solve the zero one knapsack problem using dynamic programming.
- b) Find the optimal binary search tree for the given keys and probabilities.

UNIT - 4 [Backtracking & Branch and Bound]

- a) Explain the N queens problem using backtracking.
- b) Differentiate between backtracking and branch and bound.

UNIT - 5 [NP-Hard and NP-Complete Problems]

- a) Explain NP hard and NP complete classes with examples.
- b) Explain NP hard and NP complete classes with examples.

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